Task-Relevant Subspace Motion and Risk Rebound in Quadratic Representations

Working manuscript

October 21, 2026

Abstract

When does pretraining change an independent task's linear readability, and how large or persistent can that change be? For a convolution–square–pool encoder, local second moments give an exact representation for arbitrary input distributions. Standard population regression then connects multi-output tasks to subspace geometry. We separate exterior encoder updates caused by objective/data anisotropy from those caused by parameter mobility, obtaining quantitative near-conservation for linear factorization. In a solvable quadratic family with a genuine encoder bottleneck, filter correlation and head coupling create new second-order directions. An open cone of tasks shares a fixed improvement followed by excess risk above initialization, with explicit gaps, finite arrival times, and a structured five-percent covariance range. This family also identifies a symmetry under which learning-rate ratios cannot change the ordered subspace path. More generally, initial task cross-block mass and normal motion budget bound finite risk changes, while the two projector tangents characterize which tasks admit rate competition. Source-only interval bounds give nonzero finite-window predictions on existing matched-function trajectories; saved future risks satisfy those intervals at a common energy section. Earlier certified projection rebound and population/empirical texture results provide complementary evidence. CIFAR-10 and DTD remain boundary cases: pre-existing head-selection gaps and weak class gains have not yet been explained by measured values of the new bounds. The scope is task-conditioned population geometry, not a universal learning-rate rule or a finite-sample generalization guarantee.

1 Introduction

Pretraining can improve its objective while making a representation less useful for a later task. The difficult question is structural: which updates change task readability, and what determines the size and duration of that change? Output variance, spectral concentration, and total gradient norm do not answer this question, because unregularized population linear risk depends on a subspace rather than on its coordinate scaling.

We study a convolutional encoder with squaring, global average pooling, and a linear head. Its local second-moment representation is exact for arbitrary image distributions: each channel reads $h_c^\top M(x) h_c$. Whitening the centered moment dictionary puts coordinate tasks, band-energy tasks, and arbitrary multi-output labels in one regression framework. The risk formula is standard least squares; its role is to identify the task matrix against which subspace motion must be evaluated. Independent Gaussian frequencies are a tractable special case, rather than a prerequisite for representing natural images.

The first structural distinction concerns the source of exterior motion. A linear factorization cannot move its encoder row space under an isotropic population objective. Away from isotropy, we bound its normal velocity by an explicit residual gradient. A square encoder has an additional source: anisotropy of its channel Jacobian Gram operators. For Gaussian patches these operators are Jordan products of rank-one filter matrices. With two independent filters, encoder motion requires both nonzero filter correlation and objective/head cross coupling. This criterion also supplies nonlinear cases with exact conservation.

We use that structure to solve an actual parameter flow with $K = 1 < C = 2 < d = 3$. A moving quadratic direction passes through task-aligned directions and continues beyond them. An open cone of tasks shares two comparison states at which both encoder and projection risks have improved and then exceeded initialization, with explicit uniform gaps. The family covers a continuum of scales, orientations, learning-rate ratios, and a structured five-percent covariance range. Its symmetry is a checkable sufficient initialization condition, not a high-probability statement. Within this symmetry, different rates only reparameterize the ordered subspace path, so rebound alone does not imply rate competition.

For general states we bound finite risk change by initial task cross-block mass and normal subspace movement. This separates small motion from weak task coupling as two amplitude bottlenecks. Two nonzero block projector tangents admit a competing task exactly when they are not positively proportional. Finite-budget and error bounds then specify how state drift, gradient uncertainty, and step size consume a source competition

margin. These general bounds may be conservative; they are not useful merely because their constants exist. On pre-existing matched-function population trajectories, source-only interval boxes certify finite risk changes of opposite signs at a common energy section, and the saved future values fall within their amplitude envelopes. This is a retrospective nonvacuity check with prediction constants separated from future records.

The original computer-assisted fixed-gap projection construction and controlled texture experiments are retained. They distinguish selection inside an encoder space from changes to that space, and expose systematic finite-data linear drift and late-stage prediction failure. Natural-image evidence is compressed to a boundary table, with full diagnostics in the appendix. CIFAR head-selection gaps largely predate training, and suppressing a dominant DTD statistic does not restore class learning. The new task coupling, normal budget, and error constants have not been measured there; we do not claim to explain those failures. The resulting contribution is a structural analysis with explicit constructive and numerical domains, rather than a general claim that pretraining or a particular rate ratio loses semantic information.

2 Related Work

Initialization and optimization geometry. Kunin et al. [5] analyze how layer imbalance and learning rates change feature learning, including comparisons with the same initial product. Our scale–learning-rate equivalence belongs to this broader setting. Its role here is to connect the actual parameter metric to an explicit independent-task risk threshold, rather than to establish scale sensitivity for the first time.

Projection heads and feature selection. Gui et al. [2] characterize expansion and shrinkage induced by projection heads with fixed encoders. Xue et al. [8] study layerwise feature weighting and report settings with post-projection early-stopping gains. Torralba and Weiss [7] analyze the same broad convolution–nonlinearity–GAP architecture using GUPA; their Appendix M discusses nonunique decompositions, initialization effects, and recognition performance falling while the loss improves. Our question concerns finite-time population risk along actual joint parameter training, with fixed risk gaps and explicit encoder/projection separation.

Nonlinearity and collapse. Jing et al. [3] analyze two-layer linear contrastive dynamics and dimensional collapse; Tian [6] studies the role of nonlinearity in contrastive dynamics. Chaudhry [1] develops a geometric account of projection heads. The Jacobian Gram matrix governing our effective updates is distinct from a pull-back loss Hessian. Moreover, our scalar variance remains positive and our texture experiments retain their numerical ranks: changing a task direction is not, by itself, dimensional collapse. Our contribution is scoped to the stated construction and measured models, without an exhaustive priority claim over all equivalent factorizations.

3 Model, Objective, and Risk

Images and effective coordinates. Choose d independent non-DC, non-self-conjugate Fourier pairs of a real periodic image. Let $x_k^F \sim \mathcal{CN}(0, g_k)$ independently and set

$$q_k = \frac{|x_k^F|^2 - g_k}{g_k}, \qquad \mathbb{E}q = 0, \quad \mathrm{Cov}(q) = I. \tag{1}$$

The network is $x \mapsto W\,\mathrm{GAP}((h * x)^2)$, without biases. With the spatial/Fourier normalization fixed in Appendix A, its centered encoder and projection are

$$r_{\mathrm{enc}} = Bq, \qquad z = Aq, \qquad B_{ck} = g_k |h_{ck}|^2, \quad A = WB. \tag{2}$$

For full-support kernels the Fourier coordinates are free; for local kernels they are coupled by the spatial support constraint. We retain that constraint in training and differentiation.

Pretraining objective. For centered view covariance $\Sigma = ASA^\top$ and view-difference second moment D, we analyze the Gaussian Uniformity Plus Alignment objective (GUPA) [7]

$$\begin{aligned} L(A) = {} & \tfrac{1}{2}\,\mathrm{tr}(ADA^\top) - \tfrac{1}{2}\log\det(I+\Sigma) \\ & - \tfrac{1}{2}\,\mathrm{tr}\left((I+\Sigma)^{-1}\Sigma\right). \end{aligned} \tag{3}$$

The energy features are not Gaussian embeddings; GUPA is the objective studied directly, without invoking exact InfoNCE equivalence. In the isotropic texture setting, independent spectral noise of relative power η gives $S = sI$, $D = 2(s-1)I$, $s = (1+\eta)^2$. For $K = 1$, write $A = a^\top$, $V = \|a\|^2$, and $Q = sV$. Then

$$L(a) = (s-1)V + f(sV), \tag{4}$$

$$f(Q) = -\tfrac{1}{2}\log(1+Q) - \frac{Q}{2(1+Q)}. \tag{5}$$

The theorem remains strictly inside the constraint $Q \le 1$. Empirical experiments use the full matrix objective with within-view sample centering and do not normalize individual images or outputs.

Independent tasks. For $Y = \beta^\top q + \xi$, assume $\mathbb{E}[\xi \mid q] = 0$ and $\mathbb{E}\xi^2 = \sigma^2$. The optimal population linear probe satisfies

$$R_{\mathrm{pop}}(A) = \sigma^2 + \|\beta\|^2 - \beta^\top P_A \beta, \tag{6}$$

$$P_A = A^\top (AA^\top)^\dagger A. \tag{7}$$

We denote excess risk by $\mathcal{R} = R_{\text{pop}} - \sigma^2$. Equation (6) is evaluated by an SVD row-space basis, with singular values and numerical ranks recorded. It is invariant to invertible changes of representation coordinates. For $K = 1$, $d = 2$, and $\beta = e_1$, $\mathcal{R} = a_2^2/(a_1^2 + a_2^2)$. The encoder has the same risk formula with B replacing A; an invertible B has risk σ^2 regardless of the head.

For complete orthonormal coordinate tasks, $\sum_j \mathcal{R}_{e_j}(A) = d - \operatorname{rank}(A)$. Thus fixed-rank improvements and losses represent a redistribution of a fixed total budget, not necessarily a decrease in dimension. The substantive question is which directions the training geometry selects.

4 Task-Relevant Motion in Local Moment Space

4.1 An exact bridge beyond independent frequencies

For valid patches $p_u(x) \in \mathbb{R}^p$, define

$$M(x) = |U|^{-1} \sum_u p_u(x) p_u(x)^\top. \tag{8}$$

Squaring and GAP give $r_c(x) = h_c^\top M(x) h_c$ for every input. Let $\phi = \operatorname{svec}(M - \mathbb{E}M)$, where symmetric vectorization preserves the Frobenius inner product. Write its covariance on its support as $\Sigma_\phi = U \Lambda U^\top$, $\Lambda \succ 0$, and set

$$u = \Lambda^{-1/2} U^\top \phi, \qquad \mathbb{E} u u^\top = I, \tag{9}$$

$$B_{c:} = \operatorname{svec}(h_c h_c^\top)^\top U \Lambda^{1/2}, \qquad A = WB. \tag{10}$$

Thus the centered encoder and projection are exactly Bu and Au. Gaussianity and independent frequency powers are unnecessary for this identity. They remain useful for explicit dynamics and exact moment calculations.

For any centered multi-output label Y with finite second moment, put $C_Y = \mathbb{E}[uY^\top]$ and $T = C_Y C_Y^\top$. Population affine least squares gives

$$R(A) = \operatorname{tr} \operatorname{Cov}(Y) - \operatorname{tr}(P_A T). \tag{11}$$

This standard regression identity includes coordinate, band-energy, and centered one-hot tasks without assuming that labels are linear functions of u. It describes linear readability, not all task information or finite-sample probe generalization. Below, moment-space training formulas assume augmented moments are supported on the same clean covariance support. If augmentation introduces clean-null directions, training must retain those additional coordinates; clean whitening alone does not determine its gradient.

4.2 Data anisotropy and parameter mobility are separate sources

Assume the representation matrix under discussion has full row rank on the interval. Write $P = P_A$, $Q = I - P$. For an effective update Z, its projector tangent is

$$\mathcal{D}_A[Z] = A^\dagger Z Q + Q Z^\top (A^\dagger)^\top. \tag{12}$$

Only ZQ can change its row space. If $G = \nabla_A L$, ordinary parameter training induces

$$\dot{A} = -\eta_h \mathcal{M}_h[G] - \eta_W G B^\top B, \tag{13}$$

where $\mathcal{M}_h = J_h J_h^*$ and $J_h = D_h(WB)$. Total gradient size and feature energy do not determine either projector tangent.

There is a concrete near-isotropy version of linear conservation. Decompose $G = HA + E$ and let $A = WV$ be a linear factorization. Then

$$\dot{V}(I - P_V) = -\eta_V W^\top E (I - P_V). \tag{14}$$

For GUPA let $S = sI + \Delta S$, $D = d_0 I + \Delta D$, and

$$K = (I + ASA^\top)^{-1} + (I + ASA^\top)^{-2}. \tag{15}$$

Its exact gradient is $G = AD - KAS$, so one may take

$$H = d_0 I - sK, \qquad E = A\Delta D - KA\Delta S, \tag{16}$$

$$\|E\|_F \le \|A\|_F \left(\|\Delta D\|_2 + 2\,\|\Delta S\|_2\right). \tag{17}$$

The isotropic term cannot move the encoder row space; the residual can have a systematic finite-data component. Under exact isotropy this conservation also holds for ordinary discrete gradient descent whenever the front matrix retains its rank, since each update is a left multiplication.

For the square encoder, let $J_c = D_{h_c} B_{c:}^\top$, $T_c = J_c J_c^\top = \tau_c I + \mathcal{E}_c$, and $U_0 = W^\top H W$. With $Q_B = I - P_B$, its exact exterior velocity is

$$(\dot{B} Q_B)_{c:} = -\eta_h \left[(U_0 B)_{c:} \mathcal{E}_c + (W^\top E)_{c:} T_c\right] Q_B. \tag{18}$$

This separates parameter-mobility anisotropy from objective/data anisotropy. A square parameterization does not automatically move its encoder space: scalar channel mobilities $T_c = \tau_c I$ and $E = 0$ exclude that motion. Conversely, anisotropic channel mobilities can move it even when $E = 0$. Appendix F gives a two-filter criterion: for isotropic Gaussian patches and linearly independent filters, motion occurs exactly when filter correlation and objective/head cross coupling are both nonzero.

4.3 A solvable encoder rebound family

The following family makes the mechanism explicit in a genuine bottleneck $K = 1 < C = 2 < d = 3$. Its moving directions are coupled quadratic statistics, not an embedding with dormant extra coordinates.

Let $x \sim N(0, \Sigma_x)$ be one two-dimensional patch, with

$$\Sigma_x = \nu_+ e_+ e_+^\top + \nu_- e_- e_-^\top, \qquad \nu_\pm \in [19/20, 21/20], \tag{19}$$

where e_+, e_- are orthonormal. Write $q_c = \Sigma_x^{1/2} h_c$ for the filters in whitened input coordinates, and initialize

$$q_1 = \sqrt{\epsilon}\, e_+ + \sqrt{\epsilon/2}\, e_-, \tag{20}$$

$$q_2 = \sqrt{\epsilon}\, e_+ - \sqrt{\epsilon/2}\, e_-, \qquad W = (w_0, w_0). \tag{21}$$

Set $E_\pm = e_\pm e_\pm^\top$ and $E_\times = (e_+ e_-^\top + e_- e_+^\top)/\sqrt{2}$. These give orthonormal coordinates for

$$u = \mathrm{svec}(yy^\top - I)/\sqrt{2}, \qquad y = \Sigma_x^{-1/2} x. \tag{22}$$

For tasks use the unit vector associated with the symmetric matrix

$$\beta_{\psi,z} = \sqrt{1-z^2}(\cos\psi E_+ + \sin\psi E_-) + z E_\times. \tag{23}$$

Theorem 1 (Uniform encoder and projection risk gaps)**.** *Let $0 < \epsilon \le 1/100$, $0 < w_0 \le 1/10$, $\eta_h > 0$, and $0 \le r = \eta_W/\eta_h \le 10$. Use GUPA with views $x + \xi_j$, where independent $\xi_j \sim N(0, \eta\Sigma_x)$ and $0 \le \eta \le 1/5$. Train the actual Euclidean parameters h, W. Put $\theta_0 = \arctan(1/2)$. For every task $Y = \beta_{\psi,z}^\top u$ with*

$$2\theta_0/3 \le \psi \le 5\theta_0/6, \qquad |z| \le 1/2, \tag{24}$$

the same two comparison states $t_1 < t_2$ satisfy, for both optimal encoder and projection risks,

$$R(t_1) \le R(0) - \Delta_1, \tag{25}$$

$$R(t_2) \ge R(0) + \Delta_2, \tag{26}$$

where

$$\Delta_1 = \tfrac{3}{4}\sin(\theta_0/4)\sin(\theta_0/12) > 0.00335, \tag{27}$$

$$\Delta_2 = \tfrac{3}{4}\sin(5\theta_0/6)\sin(\theta_0/6) > 0.02181. \tag{28}$$

Both states are reached before $t_2 < 5/(76\eta_h w_0^2 \epsilon)$; the loss strictly decreases and the encoder/projection ranks remain two/one. Changing r in this family changes speed and energy, but not the ordered subspace path or these gaps.

Mechanism and exact solution. Equal head entries and the displayed filter symmetry persist. Put $\lambda_\pm = \|q_1\|^2 \pm q_1^\top q_2$ and $\kappa = 2\Phi'(V)$ for the radial objective $L = \Phi(V)$, $V = \|A\|^2$. In the increasing clock $\tau = -8\eta_h \int \kappa w^2 dt$,

$$\lambda_\pm(\tau) = \frac{\lambda_\pm(0)}{1 - \nu_\pm \lambda_\pm(0)\tau}, \tag{29}$$

$$w^2 = w_0^2 + \frac{r}{4} \sum_{j \in \{+,-\}} \frac{\lambda_j - \lambda_j(0)}{\nu_j}. \tag{30}$$

The encoder retains $E_\times$ and the diagonal direction $\lambda_+ E_+ + \lambda_- E_-$; the projection retains only the latter.

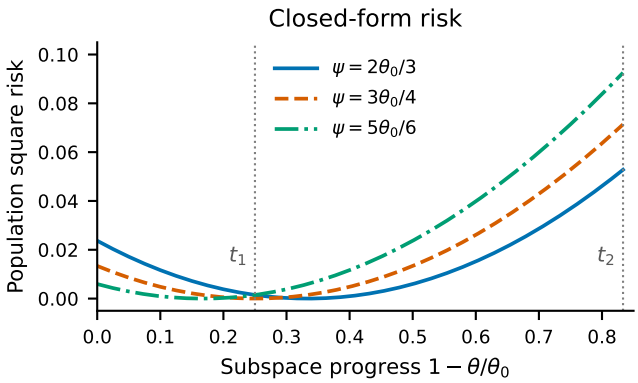


Figure 1: Exact risk paths in Theorem 1, three task directions with $z = 0$. Encoder and projection risks coincide here. The same two comparison states improve and then exceed the initial risk for the entire task family. These curves evaluate the closed formula; no model is trained or numerically integrated to draw them.

Its angle $\theta = \arctan(\lambda_-/\lambda_+)$ strictly decreases. The task risks are exactly

$$R_{\mathrm{enc}} = (1 - z^2)\sin^2(\theta - \psi), \tag{31}$$

$$R_{\mathrm{proj}} = z^2 + (1 - z^2)\sin^2(\theta - \psi). \tag{32}$$

The two common states are $\theta_1 = 3\theta_0/4$ and $\theta_2 = \theta_0/6$. Motion first approaches the task direction and then continues past it; the encoder itself loses readability. The fixed projection gap z^2 is a separate selection effect.

The task family contains an open cone and covers a specified $\pm 5\%$ covariance range. This anisotropy range keeps covariance axes aligned with filter sums/differences and uses covariance-relative augmentation and balanced whitened initialization. Physical initial filters change with the covariance to keep that whitened balance. It is not robustness to arbitrary symmetry-breaking initialization or augmentation perturbations, nor a random-initialization probability result. The family also supplies an explicit exclusion case for rate competition: unequal rates alone do not force different directional paths. Appendix F proves the formulas, energy barrier, and reachability.

4.4 How much motion can matter to a task?

Let P_0, P_1 be equal-rank projectors, $Q_0 = I - P_0$, and define

$$a_0 = \|Q_0 T P_0\|_F, \qquad d_{01} = \|P_1 - P_0\|_F/\sqrt{2}. \tag{33}$$

The task cross block a_0 and the subspace displacement constrain risk jointly:

$$|R(P_1) - R(P_0)| \le 2a_0 d_{01} + \mathrm{osc}(T) d_{01}^2, \tag{34}$$

where $\mathrm{osc}(T) = \lambda_{\max}(T) - \lambda_{\min}(T)$. This follows from the exact row/normal block decomposition in Appendix F,

rather than a fitted local slope. In particular, $a_0 = 0$ excludes a first-order change, but can permit a one-sided quadratic change. If the row block of T has eigenvalues at least $\delta > 0$ above those of the normal block and $a_0 = 0$, every departure raises risk by at least δd_{01}^2.

For a parameter path with cumulative update budget

$$b = \int (\|\dot{h}\|_F + \|\dot{W}\|_F)\, dt, \tag{35}$$

suppose a containing region has $\sigma_{\min}(A) \geq \sigma > 0$ and $\|DA[v]\|_F \leq J \|v\|$. Then $d_{01} \leq Jb/\sigma$, and Equation (34) holds with this upper bound for every rate schedule in the region. A sharper bound uses $\int \|A^\dagger \dot{A}(I - P_A)\|_F\, dt$. For the linear encoder, Equation (14) bounds that integral by

$$\int \eta_V \|V^\dagger W^\top\|_2 \|E(I - P_V)\|_F\, dt. \tag{36}$$

This is a quantitative near-conservation statement. Constants inferred from future states give an a posteriori bound; source-fixed region bounds give a forward guarantee. None of these quantities is controlled by spectral concentration alone.

4.5 Competition, duration, and an error budget

Let D_h, D_W be the two unit-rate projector tangents at a common state. A structural criterion is available before selecting a task: if either tangent vanishes, or if $D_W = \lambda D_h$ for some $\lambda > 0$, no PSD task matrix admits opposing contributions. Otherwise an open family of rank-one tasks does admit competition. This is an if-and-only-if statement; its proof uses the off-diagonal block structure of projector tangents (Appendix F). It does not put a given class-label task in that family.

To specify an actual update budget, normalize the negative block gradients to unit directions u_h, u_W and write

$$\frac{d\theta}{db} = \alpha u_h + (1 - \alpha) u_W. \tag{37}$$

The sum of block path speeds is one. Fixed α generally requires a state-dependent physical rate ratio. On a source-centered ball of radius ρ, assume $\|\nabla R\| \leq G_R$, $\|\nabla^2 R\| \leq H_R$, and $\mathrm{Lip}(u_g) \leq K$. Put $L_q = H_R + G_R K$ and $q_g = \langle \nabla R_0, u_{g,0} \rangle$. For every allocation schedule and $b \leq \rho$,

$$\begin{aligned} b \min_g q_g - \tfrac{1}{2} L_q b^2 &\leq R(b) - R_0 \\ &\leq b \max_g q_g + \tfrac{1}{2} L_q b^2. \end{aligned} \tag{38}$$

For fixed allocations with initial slopes at least m and at most $-m$, both signs persist up to $b \leq \min\{\rho, m/(2L_q)\}$; each change has magnitude at least $mb/2$. Any two schedules differ in risk by at most $b|q_h - q_W| + L_q b^2$.

Table 1: Finite changes at the first common energy section $\ell = 0.02$. Predictions use the source state only. Future entries are pre-existing numerical trajectories, not independent trials.

Scale c	Certified $R(\ell) - R(0)$	Saved change
1/2	$[0.003179, 0.007176]$	0.005326
1	$[-0.003110, -0.001259]$	−0.002280
3/2	$[-0.010741, -0.002101]$	−0.006647

With Euler steps of size at most $\delta_{\max}$ and normalized direction error at most ε, the usable margin becomes

$$m_{\text{eff}} = m - G_R \varepsilon - \tfrac{1}{2} H_R (1 + \varepsilon)^2 \delta_{\max}. \tag{39}$$

If it is nonpositive, this certificate supplies no sign guarantee. If it is positive, a sufficient cumulative window is

$$b \leq \min\left\{ \frac{\rho}{1 + \varepsilon}, \frac{m_{\text{eff}}}{L_q (1 + \varepsilon)} \right\}, \tag{40}$$

with signed change at least $m_{\text{eff}} b/2$. These are explicit sufficient bounds, not predictions that CIFAR errors are small. Appendix F supplies source-computable derivative constants and distinguishes this path budget from a fixed sum of learning rates. The generic Taylor bound is a connecting tool; its nonvacuity must be checked separately.

4.6 A nonempty forward window on existing trajectories

We test nonvacuity on three stored matched-function trajectories of the original two-frequency model. All start with $A_0 = (0.02432, 0.02432)$; the $c = 1$ source is exactly the center of Theorem 2. Source-only interval arithmetic bounds the physical vector field, risk acceleration, and first hitting time of $\ell = 0.02$. It then opens the saved trajectory CSV for validation, without integrating a new trajectory. Table 1 gives conservative outward display bounds.

The source bounds certify nonzero changes with opposite signs at the same initial function and common progress section; the first two have risk separation greater than 0.00443. All saved hitting times and risks fall within their predicted intervals. The physical windows also contain five saved future records, all inside the amplitude envelopes. This is a retrospective check of a forward certificate, not a newly preregistered experiment. It supplies a finite-window counterpart to the local threshold on this construction family, not an independent validation of every general moment-space bound or a natural-image prediction. Appendix F.7 gives the boxes, failed shorter energy-clock bounds, and the separate fixed-rate-sum comparison.

5 Certified Projection Rebound and Matched-Function Rates

5.1 A finite-time fixed-gap construction

Take $C = d = 2$, $K = 1$, $\beta = e_1$, and unit parameter learning rates. Define the positive initialization coordinates

$$x = (w_1, w_2, B_{11}, B_{12}, B_{21}, B_{22}, g_1, g_2) \tag{41}$$

and the rational center

$$\bar{x} = (1/25, 9/25, 8/25, 4/125, 4/125, 8/125, 1, 2). \tag{42}$$

Each B_{ck} is realized by $|h_{ck}|^2 = B_{ck}/g_k$. Spectral powers g are fixed along each trajectory, but vary across the initialization box. Set $\ell = \log(V/V_0)$ separately for each trajectory.

Theorem 2 (Certified fixed risk gaps)**.** *Let $\delta = 10^{-5}$ and let $x_i(0) \in [(1-\delta)\bar{x}_i, (1+\delta)\bar{x}_i]$ independently. For every $\eta \in [0, 1/5]$, Euclidean parameter gradient flow for the isotropic GUPA objective reaches $\ell_1 = 1.7$ and $\ell_2 = 6$ in finite physical time. Throughout this interval the parameters and vector-field denominators remain positive and $Q < 0.689 < 1$. Moreover,*

$$\mathcal{R}(1.7) \le \mathcal{R}(0) - 0.04, \tag{43}$$
$$\mathcal{R}(6) \ge \mathcal{R}(0) + 0.15, \tag{44}$$

and hence $\mathcal{R}(6) - \mathcal{R}(1.7) \ge 0.19$. The pretraining loss strictly decreases between these times and initialization.

The relative half-width is 0.001%, not 1%. This is a continuous-box existence result, not a high-probability random-initialization theorem. The guarantee concerns two sections and does not assert a unique risk minimum or monotonicity between sections. It does not assert encoder deterioration: whenever the two-dimensional encoder is invertible, its optimal encoder risk is at the noise floor. The theorem concerns the compressed projection.

Proof structure. Write $\gamma(Q) = (Q+2)/(2(Q+1)^2)$ and $\Gamma_s(Q) = s\gamma(Q) - (s-1)$. A positive time change $d\zeta/dt = 2\Gamma_s(Q)$ gives

$$w_c' = \sum_k B_{ck} a_k, \qquad B_{ck}' = 4 g_k B_{ck} w_c a_k. \tag{45}$$

For $K_B = B^\top B + \mathrm{diag}(4g_k \sum_c B_{ck} w_c^2)$ and $H = a^\top K_B a$, $d\ell/d\zeta = 2H/V > 0$. The resulting rational vector field is integrated with outward-rounded fixed-point intervals. A Taylor enclosure certifies the center trajectory, and a logarithmic-coordinate perturbation bound encloses every initialization in the box. Strict positivity, the energy bound, and $\Gamma_s \ge 0.1$ ensure finite physical reachability. Appendix B supplies the enclosure argument and exact reproduction settings.

The stored certificate is stronger than the displayed conservative theorem: it gives improvement at least 0.09122, excess over initialization at least 0.33664, and rebound at least 0.42790. These are certified bounds, not sampled means.

5.2 Matched functions can move in opposite task directions

Proposition 3 (Scale and learning-rate equivalence)**.** *For a loss depending on $A = WB(h)$ with $B(ch) = c^2 B(h)$, scaling $h \mapsto ch$ and $W \mapsto c^{-2}W$ preserves the initial function. Euclidean gradient training at this scaled initialization is equivalent, in the unscaled coordinates, to learning rates*

$$(\eta_h, \eta_W) \mapsto (c^{-2}\eta_h, c^4 \eta_W). \tag{46}$$

This holds step by step for ordinary gradient descent. The relative rate acquires the factor c^6.

Proof. In coordinates $h_c = c\bar{h}$, $W_c = c^{-2}\bar{W}$, the gradients scale as $\nabla_{h_c} L = c^{-1}\nabla_{\bar{h}} L$ and $\nabla_{W_c} L = c^2 \nabla_{\bar{W}} L$. Substituting into the update gives the stated factors. □

For transparent threshold arithmetic, display the unshrunk representative of the same construction family:

$$B_0 = \begin{pmatrix} 2 & 1/5 \\ 1/5 & 2/5 \end{pmatrix}, \quad w_0 = \begin{pmatrix} 1/10 \\ 9/10 \end{pmatrix}, \quad g = (1, 2). \tag{47}$$

All scaled versions have $a_0 = (19/50, 19/50)^\top$. For $S = I, D = 0$, direct differentiation yields

$$\left.\frac{d\mathcal{R}}{d\ell}\right|_0 = \frac{235 - 480c^6}{834 + 1300c^6}. \tag{48}$$

The sign reverses at $c_* = (47/96)^{1/6} \simeq 0.887778$, or relative learning rate $r_* = 47/96$ in the unscaled coordinates. Opposite derivatives at the same initial representation and equal-energy section cannot be removed by a positive time change. This threshold is specific to the construction, not a universal recommended ratio.

Multiplying all physical parameters h, W by $\rho = 2/5$ gives exactly the center of Theorem 2: $B \mapsto \rho^2 B$, $W \mapsto \rho W$. For a descending radial objective, this uniform scaling preserves the row-space path in relative-energy coordinates. The threshold therefore also holds at the certificate center, and Section 4.6 certifies its finite-window rate split there. This connects the local threshold and fixed-gap example within one family. It does not transfer physical reachability to the unshrunk state: that state may leave the descending regime before $\ell = 6$.

5.3 Which space changes?

Proposition 4 (Linear population row-space conservation)**.** *Suppose $A = WV$, $L(A) = F(AA^\top)$, and $G = \nabla_A L = H(A)A$, with H continuous along a finite,*

bounded parameter trajectory. Then ordinary linear gradient flow preserves $\operatorname{row} V(t) = \operatorname{row} V(0)$. *Every task's optimal population encoder risk is constant.*

Proof. The front-layer update is

$$\dot{V} = -\eta_V W^\top H(A) W V = C(t) V. \tag{49}$$

Its solution is $V(t) = U(t)V(0)$, where $U' = C(t)U$, $U(0) = I$, and $\det U(t) = \exp(\int_0^t \operatorname{tr} C(s)ds) \neq 0$. Left multiplication preserves row space; Equation (6) completes the proof. □

This statement covers matrix outputs, not just $K = 1$. The front-layer contribution to $\dot{A}$ is also a left multiplication of A, so its first-order contribution to row-space risk is zero. At a fixed state, changing a positive rate ratio cannot reverse this linear model's instantaneous population risk sign; finite horizons can still differ through later states.

For the square model, the actual full-support parameter update instead gives

$$\dot{B}_{ck} = -4\eta_h g_k B_{ck} (W^\top G)_{ck}. \tag{50}$$

There is generally no common left multiplier of B. In a bottleneck $C < d$, its encoder row space can therefore move. This does not imply that movement is always harmful or necessary for projection risk rebound. Linear factorization itself can produce substantial projection motion within its conserved encoder space.

5.4 Task-conditioned local interventions

For a full-row-rank representation matrix X, let $v = (XX^\top)^{-1} X\beta$ and $e = (I - P_X)\beta$. Differentiation of the least-squares residual gives

$$\dot{\mathcal{R}} = -2v^\top \dot{X} e. \tag{51}$$

Separating parameter groups yields $\dot{\mathcal{R}} = \eta_h c_h + \eta_W c_W$, where $c_h = -\langle \nabla_h R, \nabla_h L \rangle$ and likewise for W. Opposite signs permit a positive critical ratio $r_* = -c_h/c_W$; equal nonzero signs exclude immediate reversal by positive rates. We calculate coefficients from effective maps and update geometry before executing branches. The rule uses known task direction β and is an oracle mechanism test, not a label-free controller or a guarantee of long-horizon benefit.

6 Controlled Texture Experiments

6.1 Protocol and numerical checks

We generate 32×32 periodic textures using four directions, two radial bands per direction, and two independent conjugate frequency pairs per group, for $d = 16$ independent power coordinates. The initial spectrum is flat. Images remain floating point without per-image normalization or clipping. Pretraining uses no downstream labels. Ten predefined unit-variance tasks comprise two individual power coordinates and eight direction/band energy sums; all are evaluated on the same checkpoints.

The main bottleneck has $C = 8$, $K = 1$; extensions use $K = 2$ or 11×11 periodic kernels. Parameters have no bias, batch normalization, weight decay, momentum, or output normalization. We compare $r = \eta_W/\eta_h \in \{0.1, 1, 10\}$. Each seed shares initial effective representations and paired data/augmentation streams across ratios. Controls are direct effective-map optimization, ordinary linear factorization on the same q, and frozen layers. A linear control on raw pixels would change task expressivity and is not used.

We retain two calibration protocols. The matched-parameter protocol starts from the same B_0, W_0. Independent calibration instead uses each model's parameter Jacobian Gram traces, without task labels, to equalize its two active blocks and normalize initial total mobility. Scalar factor rescaling preserves A_0 and encoder row space. Equal traces do not imply equal update directions. Development seeds 0 and 1 are separate from ten confirmation seeds 2–11. Population protocols were locked before confirmation; settings reuse the same ten seeds and do not supply independent repetitions.

Spatial convolution, FFT evaluation, and effective-map gradients are checked independently. Centered spatial output agrees with Aq to approximately 9.71×10^{-17}, and the spatial/FFT gradient discrepancy is approximately 6.94×10^{-18}. The real-parameter fixed-section error at $\ell = 6$ decreases from approximately 4.02×10^{-4} to 2.01×10^{-4} when the step is halved. An independent empirical gradient check has approximately 1.98% relative error, reflecting moment estimation rather than machine-precision equivalence. No finite-sample trajectory is treated as a pointwise theorem reproduction.

6.2 Population subspace selection

The confirmation suite contains three architectures, two calibrations, three models, three ratios, and ten paired seeds: 540 population trajectories. Its primary comparison is the common section $\ell = 3$. Let $D_X = \left\| P_{X(t)} - P_{X(0)} \right\|_F$. Table 2 reports independently calibrated $r = 1$ means.

Every observed square-model seed has resolvable encoder motion, whereas linear encoder displacement remains at rounding scale. Frozen-front displacement is zero. All observed encoder and projection ranks remain 8 and K. Changing r also changes square-model displacement: for $K = 1$ full support, its mean is 0.7161, 0.3064, and 0.0485 for the three ratios. These are geometric changes, not uniformly worse task risks. Direct optimization of a scalar effective map under the radial population objective preserves its direction, while ordinary linear

Table 2: Population subspace displacement at $\ell = 3$, independently calibrated $r = 1$, ten paired seeds. Linear encoder conservation coexists with projection motion.

	Encoder D_B		Projection D_A	
Setting	Square	Linear	Square	Linear
$K = 1$, full	0.3064	$1.15\,10^{-14}$	0.5161	0.6087
$K = 2$, full	0.4499	$1.23\,10^{-14}$	0.5597	0.5319
$K = 1$, local	0.3786	$1.26\,10^{-14}$	0.5572	0.6196

Table 3: Encoder displacement after complete empirical training, evaluated at common $\ell = 1.5$; ten-seed means. Frozen-front values are zero.

Kernel	Model	$r = 0.1$	$r = 1$	$r = 10$
Full	Square	0.41312	0.13692	0.01843
Full	Linear	0.02336	0.00666	0.00089
Local	Square	0.44250	0.17740	0.02626
Local	Linear	0.02851	0.00920	0.00123

factorization can change projection risk. Rebound is therefore not unique to the square architecture.

6.3 Complete empirical training and reproducible drift

We train from initialization on 32,768 unlabeled images, batch size 256, for 4,000 updates. Full-support and local-kernel suites each contain 90 trajectories: three models, three ratios, and ten seeds. All reach the prespecified common section $\ell = 1.5$. Forty additional trajectories provide three independent SGD streams per seed for the square and linear models at the repeated condition. Table 3 reports encoder displacement.

Empirical square-model geometry remains close to the corresponding population state: at $r = 1$, $\ell = 1.5$, the empirical/population encoder projector distance averages 0.00713 for full support and 0.00850 for local kernels, compared with movements of 0.13692 and 0.17740 from initialization. This agreement concerns this model family, dataset size, and training budget.

Linear encoder movement is nonzero and reproducible. Conditional on initialization and the finite dataset, the mean pairwise cosine between terminal projector changes from independent streams is 0.9810 for the square model and 0.8683 for the linear model. These are within-seed agreements, not evidence for a universal direction across initializations. Finite datasets need not have isotropic power covariance, so their conditional expected gradient need not satisfy Proposition 4. Exploratory conditional-moment calculations on two development states support this explanation. Nonlinear sample-moment estimation and finite update size may also contribute; the diagnostic is not an independently confirmed decomposition of all drift.

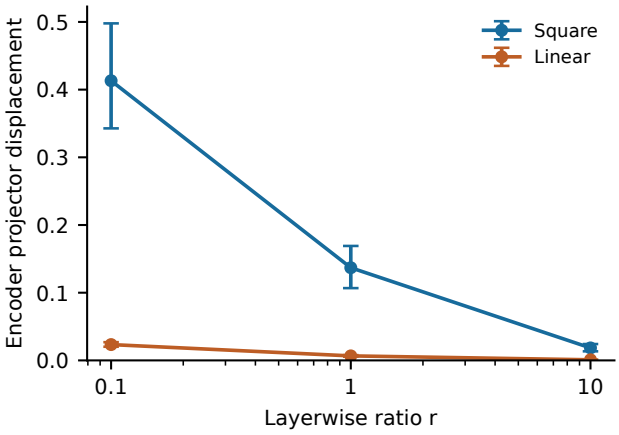


Figure 2: Full-support empirical encoder displacement at common $\ell = 1.5$. Error bars are descriptive paired-seed bootstrap intervals from the saved summary. Square-model motion is larger, but finite-data linear motion is nonzero.

6.4 Intervention prediction has a training-stage boundary

The final test uses checkpoints at updates 500, 2,000, and 3,500 from complete empirical training. Each of 90 model/seed/checkpoint states has 32 independent future batch/augmentation streams. At the midpoint, eight streams continue for 8 and 64 updates. Population geometry predicts task-conditioned risk changes before drawing future batches; actual branches use empirical GUPA updates. Seed averages, rather than branches or tasks, determine summaries.

For the square model, single-step sign agreement is 91.8%, 57.0%, and 52.6% at the three stages; constant-decrease baselines are 55.6%, 47.7%, and 50.8%. At the midpoint the sign of the repeated-batch mean agrees 92.3% of the time even though individual batches agree only 57.0%. Late agreement is close to baseline for both summaries. This distinguishes prediction of an average direction from reliable control of a single stochastic update.

Longer midpoint windows raise sign agreement to 69.8% and 71.0%, but amplitude mean absolute error rises from approximately 1.2×10^{-5} for one update to 6.7×10^{-4} for 64 updates. Noise averaging can improve signs while the first-order magnitude approximation worsens. For the prespecified primary task, positive population thresholds occur in 5/10 early states and only 1/10 at each later stage. Threshold errors are reported only where comparable positive thresholds exist; we do not force every state to admit a reversal. The intervention remains an oracle mechanism experiment using known β.

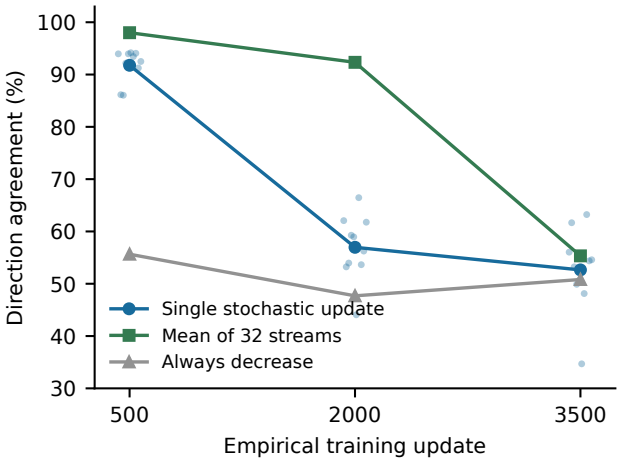


Figure 3: Square-model one-update direction predictions at empirical checkpoints. Markers average ten seed-level summaries; thin points show seeds. The distinction between individual stochastic updates and replicate means disappears late.

Table 4: Natural-image boundary evidence: validation OLS accuracy (%), initialization → final budget. Three development seeds per row; CIFAR frozen readouts share the same encoders. Rank-8 fitting uses training labels. DTD uses official partition 1.

Dataset	Representation	Initial	Final
CIFAR	Encoder	36.31	36.04
	Original eight-dimensional head	23.49	23.71
	Supervised rank-8 readout	35.75	35.59
DTD	Encoder	8.65	8.51
	Zero-DC encoder	8.78	8.56

7 Natural Images as Boundary Evidence

The local second-moment representation in Section 4 is an architecture identity for these experiments. Its population risk and movement bounds do not, however, predict natural-image behavior until their task moments, normal update budgets, and stability constants have been measured. The existing CIFAR-10 and DTD results delimit this gap; they are not a validation of the new finite-horizon bounds. Full protocols, capacity scans, DC interventions, and local derivative diagnostics are retained in Appendix G.

CIFAR uses grayscale images, 64 local 11×11 filters, squaring, GAP, and an eight-dimensional head. At a fixed frozen encoder, supervised rank-8 least squares recovers almost all of the full encoder's validation MSE advantage over the original head. The approximately twelve-point accuracy gap is already present at initialization (Table 4). It demonstrates available class-dependent compression, not a large training-induced deletion of class information. Reduced-rank fitting optimizes training MSE, not validation accuracy. The separate five-seed final-test comparison has encoder accuracy near 36.5% across rate ratios; equal-energy validation comparisons do not establish a stable speed-independent gain.

On DTD, objective improvement concentrates encoder variance in an existing image statistic associated with DC response. Removing that response at evaluation or training with spatially zero-mean filters does not yield clear encoder class gains. Zero-DC training instead emphasizes a statistic associated with pixel variance. These outcomes are consistent with objective–task mismatch, but neither spectral concentration nor removal of one dominant statistic identifies the cause of weak class learning.

The existing label-conditioned CIFAR derivative diagnostic has eligible opposing contributions in only four of 27 state–stream combinations. All four reverse the first-step risk sign, with changes of order 10^{-9}; none preserves both predicted signs at twenty updates. Predictions and the first update share the same gradient and use validation labels. This is a local calculation check, not an independent finite-horizon test. Source-state task coupling, separate future-gradient errors, and usable curvature bounds have not been measured for these runs. We therefore do not claim that the new theory explains their small magnitude or short persistence. Testing that claim requires calibration states or batches separated from the future update stream, rather than further searches for classification gains.

8 Discussion and Limitations

The structural question has two parts: whether a training update leaves the current representation subspace, and whether that exterior movement intersects task-predictive directions. The local moment identity and regression formula make both parts explicit without assuming independent frequencies. The near-conservation decomposition separates data/objective anisotropy from parameter-mobility anisotropy, while the finite displacement bound identifies task cross-block mass and normal movement as separate limits on risk amplitude.

The solvable bottleneck family explains rebound as passage through a task-aligned direction, followed by continued movement. Its fixed gaps hold for both encoder and projection, across an open task cone and a specified covariance interval. These are substantive extensions of a single compressed-output example. They remain conditional: equal head entries, balanced whitened filters, covariance-axis alignment, and relative-covariance augmentation define the invariant family. The five-percent range is structured anisotropy, not arbitrary perturbation stability. No random-initialization occurrence probability or general stability radius away from this symmetry is

established. Moreover, this family excludes rate-induced path competition; that mechanism is diagnosed by the separate two-tangent criterion.

The finite-window inequalities provide a falsifiable certificate rather than a promise of useful control. A small task cross block suppresses first-order transfer; small normal movement limits total transfer; a small competition margin can be overwhelmed by drift, gradient error, or discrete curvature. Generic source bounds can be empty or too conservative. The interval calculation is nonempty on a stored positive construction and predicts later changes at a common energy section. It uses a known population task, a deterministic vector field, and short windows. It does not establish useful path-budget constants for arbitrary states, stochastic optimization, or natural-image OLS. The analysis was designed after the trajectories existed, with constants calculated from source states before future CSV records were read.

The earlier narrow-box certificate remains a computer-assisted proof of a larger projection-risk rebound, not an encoder-loss theorem or a probability statement. The scale equivalence belongs to established work on initialization imbalance. The generic regression, projector calculus, and Taylor remainder are connecting tools; their presence alone is not a novelty claim. A theorem-level comparison of the new constructive family and task-geometry criteria with the closest literature remains necessary before a submission-level contribution judgment.

Empirical textures retain a clear conditional correspondence and a boundary: population geometry is informative early, but finite data induce reproducible linear drift and late predictions weaken. Under exact isotropy, ordinary discrete linear gradient descent also preserves its encoder row space as long as rank is retained, so step size alone cannot explain that drift. The existing empirical objectives and moments need not satisfy this exact condition.

CIFAR-10 and DTD remain useful negative evidence. Better supervised low-rank heads expose class information already present at initialization; dominant-statistic removal does not yield clear encoder class gains; rare local risk reversals are tiny and short-lived. None of those observations by itself identifies the new bound's controlling quantity. A direct natural-image test would need source-calibrated task coupling and motion/error bounds checked against separate future states or update streams. We have not run that test, and have not resumed large-scale training. Weak class gains therefore remain an unresolved transfer boundary, not an explained success of the theory.

9 Conclusion

Task risk changes through exterior subspace motion coupled to the task matrix. We give a local-moment decomposition of that motion, an analytic encoder-bottleneck rebound family with explicit gaps and a structured anisotropy range, and finite amplitude and window criteria. A source-only interval check predicts nonzero opposite risk changes on existing matched-function trajectories. The retained texture and natural-image results distinguish established mechanisms from the conditions that still lack predictive validation.

References

[1] Faris Chaudhry. The geometry of projection heads: Conditioning, invariance, and collapse. *arXiv preprint arXiv:2605.17180*, 2026. URL https://arxiv.org/abs/2605.17180v1. Version 1.

[2] Yu Gui, Cong Ma, and Yiqiao Zhong. Unraveling projection heads in contrastive learning: Insights from expansion and shrinkage. *arXiv preprint arXiv:2306.03335*, 2023. URL https://arxiv.org/abs/2306.03335v1. Version 1.

[3] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2110.09348v3.

[4] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

[5] Daniel Kunin, Allan Raventós, Clémentine Dominé, Feng Chen, David Klindt, Andrew Saxe, and Surya Ganguli. Get rich quick: exact solutions reveal how unbalanced initializations promote rapid feature learning. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/94074dd5a072d28ff75a76dabed43767-Abstract-Conference.html.

[6] Yuandong Tian. Understanding the role of nonlinearity in training dynamics of contrastive learning. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2206.01342v3.

[7] Antonio Torralba and Yair Weiss. A theory of contrastive learning with natural images. *arXiv preprint arXiv:2607.07470*, 2026. URL https://arxiv.org/abs/2607.07470v1. Version 1.

[8] Yihao Xue, Eric Gan, Jiayi Ni, Siddharth Joshi, and Baharan Mirzasoleiman. Investigating the benefits of projection head for representation learning. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2403.11391v1.

A Spatial Metric and Effective Updates

For image side length n, use orthonormal real cosine/sine basis functions $\sqrt{2/n^2}\cos(2\pi k\cdot x/n)$ and the corresponding sine functions. Circular correlation is divided by $\sqrt{n^2/2}$. The physical spatial kernel then has Euclidean coordinates $h_{ck,\cos}, h_{ck,\sin}$, and

$$B_{ck} = g_k(h_{ck,\cos}^2 + h_{ck,\sin}^2). \tag{52}$$

This convention counts each independent conjugate pair once. The effective centered map is obtained by subtracting its distributional mean; empirical GUPA uses sample centering and probes include an intercept. Differentiating physical kernels gives Equation (50), including its factor of four. Fourier SGD with a different normalization is not automatically the same algorithm.

For a local kernel let $J_c = D_{h_c}B_{c:}$ and $T_c = J_cJ_c^\top$. Then, in column-vector notation,

$$\dot{B}_{c:}^\top = -\eta_h T_c(W^\top G)_{c:}^\top. \tag{53}$$

For full support $T_c = \mathrm{diag}(4g_kB_{ck})$; local support introduces cross-frequency terms. These terms are retained in the experiments. Effective updates are parameter Jacobian Gram updates, not arbitrary effective-map optimization.

B Details of the Certified Fixed-Gap Theorem

B.1 Energy-parameterized rational dynamics

Let $a = B^\top w$, $V = a^\top a$, $K_B = B^\top B + \mathrm{diag}(4g_k\sum_c B_{ck}w_c^2)$, and $H = a^\top K_B a$. In effective time, Equation (45) gives $a' = K_Ba$ and $V' = 2H$. Positivity implies $H > 0$. In $\ell = \log(V/V_0)$ coordinates, the eight-dimensional vector field is

$$F_{w_c}(x) = \frac{V}{2H}\sum_k B_{ck}a_k, \tag{54}$$

$$F_{B_{ck}}(x) = \frac{V}{2H}4g_kB_{ck}w_ca_k, \tag{55}$$

$$F_{g_k}(x) = 0. \tag{56}$$

Along exact solutions $dV/d\ell = V$, hence $V(\ell) = V_0e^\ell$. Although each initialization has its own V_0, the rational field contains no external V_0 parameter. This permits a common-section comparison across the box.

B.2 Outward-rounded trajectory enclosures

The certificate uses step $h = 1/500$, 3,000 steps, and fixed-point endpoints with denominator 2^{180}. The two sections are exactly steps 850 and 3,000. All proof decisions use integer outward rounding; floating point is used only for readable progress output. Any denominator containing zero causes failure.

For each center-trajectory endpoint enclosure X_n, the verifier constructs a positive trajectory tube T_n and checks strict Picard inclusion in the six dynamic coordinates:

$$X_n + [0,h]F(T_n) \subset \mathrm{int}(T_n). \tag{57}$$

The two static coordinates retain their input intervals. A first-exit argument establishes existence and containment throughout the step. Truncated power-series arithmetic computes coefficients recursively,

$$c_0 = x, \qquad c_{j+1} = \frac{[z^j]F(\sum_{k=0}^j c_kz^k)}{j+1}. \tag{58}$$

The step endpoint satisfies the interval Taylor inclusion

$$x(\ell_n + h) \in \sum_{j=0}^{7} c_j(X_n)h^j + c_8(T_n)h^8. \tag{59}$$

Thus the enclosure includes the eighth-order remainder, all arithmetic rounding, and the entire within-step trajectory. Checking only endpoints would not suffice.

B.3 Enclosing the continuous initialization box

Directly propagating an initial box can inflate intervals. Instead, enclose the center and control neighboring solutions in logarithmic coordinates $y = \log x$. Define $f_i(y) = F_i(e^y)e^{-y_i}$ and $J = D_yf$. For the initial relative half-width $\delta = 10^{-5}$,

$$\|y(0) - \bar{y}(0)\|_2 \le \sqrt{8}\frac{\delta}{1-\delta} < \frac{3}{99999} = E_0. \tag{60}$$

For each step the verifier certifies

$$\lambda_{\max}\big((J + J^\top)/2\big) < m_n \tag{61}$$

throughout the inflated trajectory tube. It searches $m_n \in \{1/20, 2/20, \ldots, 80/20\}$ and sets, with upward rounding,

$$r_n = \frac{E_n}{1 - m_nh}, \qquad E_{n+1} = r_n < 1/2. \tag{62}$$

Since $e^{m_nh} < 1/(1-m_nh)$, this bounds the whole step. The required tube is the componentwise product

$$\widehat{T}_n = T_n[1 - r_n, (1-r_n)^{-1}]. \tag{63}$$

It contains exponentials of the log-error ball and the log-coordinate segments joining neighboring solutions. The derivative is evaluated without approximate logarithms:

$$J_{ij} = \frac{x_j}{x_i}\frac{\partial F_i}{\partial x_j} - \mathbf{1}_{i=j}\frac{F_i}{x_i}. \tag{64}$$

The static g rows are zero, but their influence on other rows remains included.

To prove Equation (61), let $\mathscr{S}$ enclose the symmetric part of J, let M be its symmetric fixed-point midpoint matrix, and let e_n bound the maximum absolute row sum of interval deviations from M. Every real $S \in \mathscr{S}$ obeys $\|S - M\|_2 \le e_n$. The verifier checks

$$(m_n - e_n)I - M \succ 0 \tag{65}$$

by outward-rounded unpivoted $LDL^\top$ recursion with all eight pivots strictly positive. Failure of every candidate m_n aborts verification. Integrating the symmetric-Jacobian bound along segments and using a first-exit argument closes the error tube. No approximate eigenvalue is accepted as a proof decision.

B.4 Risk bounds, constraints, and physical time

At a section, inflate X_n by $[1 - E_n, (1 - E_n)^{-1}]$. For $a_1 \in [A_-, A_+]$, $a_2 \in [C_-, C_+]$ with positive endpoints, use

$$\mathcal{R} \in \left[\frac{C_-^2}{A_+^2 + C_-^2}, \frac{C_+^2}{A_-^2 + C_+^2}\right]. \tag{66}$$

Subtracting section and initialization intervals gives uniform bounds for each initialization, conservatively ignoring their correlations. Readable outward bounds are

$$0.49997 \le \mathcal{R}(0) \le 0.50003, \tag{67}$$

$$0.40844 \le \mathcal{R}(1.7) \le 0.40876, \tag{68}$$

$$0.83666 \le \mathcal{R}(6) \le 0.83883. \tag{69}$$

The stronger gap constants quoted in the main text use the unrounded binary-rational endpoints, not subtraction of these rounded displays.

The energy bound holds continuously, using $\bar{V}_0 = 0.0011829248$ and an exact rational series proof of $e^6 < 404$:

$$Q(\ell) \le \frac{36}{25}(1 + \delta)^4 \bar{V}_0 e^6 < 0.689. \tag{70}$$

For $1 \le s \le 36/25$ and $Q \le 1$, $\Gamma_s(Q) \ge 0.1$. Therefore

$$\frac{d\ell}{dt} = 4\Gamma_s(Q)\frac{H}{V} > 0, \qquad \frac{dL}{d\ell} = -\Gamma_s(sV)V < 0. \tag{71}$$

The compact positive tube has a uniform lower bound on H and an upper bound on V, implying finite $dt/d\ell$ and finite physical reachability. This completes the computer-assisted proof of Theorem 2, conditional on the explicitly checkable arithmetic and inclusion tests.

A matched-function subfamily. Fix $g = (1, 2)$ at the center and scale $B_0 = c^2\bar{B}$, $w_0 = c^{-2}\bar{w}$ with $c \in [1 - 1/400000, 1 + 1/400000]$. Rational endpoint checks give $c^2, c^{-2} \in [1 - \delta, 1 + \delta]$. This whole family lies in the certified box and has the same $a_0 = (0.02432, 0.02432)^\top$ and $\mathcal{R}(0) = 1/2$. The full box itself does not match a single initial representation.

Reproduction. From the repository root, run `python -B` on `experiments/code/theory/verify.py` and `experiments/code/theory/check_arithmetic.py`, with assertions enabled. The verifier reconstructs the certificate from rational initialization without reading the stored certificate. The independent Fraction-based arithmetic check does not replace the ODE enclosure. The proof is not formalized in Lean or Coq.

C Derivation of the Direction Threshold

Let $P = B^\top B$, $F_B = \mathrm{diag}(4g_k \sum_c B_{ck} w_c^2)$, and $K_B = P + F_B$. For the matched-function initialization in Section 5,

$$P_0 = \begin{pmatrix} 101/25 & 12/25 \\ 12/25 & 1/5 \end{pmatrix}, \quad F_0 = \mathrm{diag}(91/125, 326/125). \tag{72}$$

Scaling gives $K_B = c^4 P_0 + c^{-2} F_0$. With $a_* = 19/50$ and $a_\perp = (-a_2, a_1)^\top$,

$$N_0 = a_\perp^\top K_B a = a_*^2 c^{-2}(47 - 96c^6)/25, \tag{73}$$

$$a_0^\top K_B a_0 = a_*^2\big((26/5)c^4 + (417/125)c^{-2}\big). \tag{74}$$

At equal positive coordinates, the derivative of $\mathcal{R} = \sin^2\vartheta$ with respect to angle is one. Thus $d\mathcal{R}/d\ell = N_0/(2a_0^\top K_B a_0)$, which simplifies to Equation (48). The threshold changes only the initial sign. The main fixed-gap theorem does not infer a long-time trajectory from this local calculation.

More generally, any smooth radial objective $\Phi(V)$ with $-\Phi'(V)$ bounded below by a positive constant on the trajectory interval shares Equation (45) after a positive time change. The energy-parameterized paths are consequently independent of the particular radial loss on that interval. Anisotropic objectives do not generally have this scalar-factor property.

The threshold representative and certificate center are related by uniform physical scaling $\rho = 2/5$, which sends (w, B, a, V, K_B, H) to $(\rho w, \rho^2 B, \rho^3 a, \rho^6 V, \rho^4 K_B, \rho^{10} H)$. In Equation (56), the w components therefore scale by ρ and the B components by ρ^2. The relative-energy flow is equivariant under this scaling, so $R(\ell)$ and the initial threshold agree on common reachable sections. Positivity of the radial speed and physical reachability remain separate hypotheses, certified through $\ell = 6$ only for the stated small-energy box.

D Finite-Label Probes and Reduced-Rank Fitting

Equation (6) follows by minimizing $\sigma^2 + \left\|\beta - A^\top v\right\|^2$ over v. An invertible feature coordinate transformation

preserves the feasible prediction space. For a full-column-rank finite design and an invertible encoder, unregularized affine OLS is similarly unchanged in exact arithmetic. Fewer labels alone do not prove encoder deterioration; minimum-norm conventions in underdetermined settings, regularization, or numerical truncation require separate analysis.

For the frozen-encoder experiment, center and invertibly column-standardize the training feature matrix to obtain X, and center training one-hot targets to obtain Y. Write the numerical-rank SVD as $X = USV^\top$. Orthogonal decomposition gives

$$\|Y - XM\|_F^2 = \left\|(I - UU^\top)Y\right\|_F^2 + \left\|U^\top Y - SV^\top M\right\|_F^2. \tag{75}$$

The best rank-K approximation of $U^\top Y$ is its truncated SVD. Mapping back through VS^{-1} gives a rank-K linear feature projection and decoder; centering is absorbed in decoder bias. This globally minimizes training square error over rank-K affine predictors, without using validation labels. It is not globally optimal for classification accuracy or for the CE training objective. The source implementation additionally refits an OLS decoder to the resulting projection, enabling an independent consistency check.

E Experimental Scope and Reporting

Population suites contain 540 trajectories, complete empirical suites 180 primary trajectories, and random-stream repeats 40 additional trajectories. These counts are computational workloads. The texture inferential unit remains ten paired seeds. Repeated checkpoints, ten tasks, and future batch streams are correlated measurements. CIFAR fixed-budget comparison uses five paired seeds; capacity and frozen-head diagnostics use three development seeds, not new confirmation samples.

The theoretical and population sections use exact known-distribution risks. Texture empirical training still evaluates those risks analytically rather than fitting them on test labels. Separate pilot finite-label OLS uses nested label counts 256, 1,024, and 8,192; it is not part of the proof. CIFAR OLS fits on the existing 45,000-image training split; DTD OLS fits on the official partition-1 training set and evaluates on its validation set. Only the fixed-budget five-seed comparison uses the standard test set; the later capacity and frozen-head diagnostics use validation only.

For the texture interventions, predicted changes with absolute magnitude at most 10^{-9} are treated as numerically neutral and excluded from sign summaries. Positive-threshold existence and conditional threshold errors are reported separately. Early and late states cannot be pooled into a single reliability claim. Population-trained checkpoint branching is preliminary evidence; the final intervention table uses states reached by complete empirical training.

The CIFAR supervised references use Adam, front/head learning rate 3×10^{-4}, classifier learning rate 10^{-3}, and prescribed global gradient clipping at 1. Fixed initial feature standardization is used only in their classifier path. These choices preserve the represented function family but change optimization relative to unsupervised SGD. The local supervised stability tests triggered clipping on approximately 98–99% of updates; those local counts do not independently certify clipping rates for server runs. Current server capacity and frozen-head evidence was imported as summaries; raw server checkpoints and optimization logs were not re-audited locally.

Natural-image local directions. The CIFAR direction experiment reuses the existing training/validation split and source initialization mobility, with base learning rate 0.002 and batch size 256. The first update averages four minibatch gradients, shared by predictions and all branches from the same state; later updates use single-minibatch SGD. Future batches and augmentations are paired between branches. Model, objective, and OLS arithmetic use float64, with CPU float32 augmentation; this is a new experiment from saved weights, not exact continuation of the original float32 trajectory. Validation-risk differentiation propagates through the full-rank training OLS solve, including feature centering and scaling. Two central differences along each base-rate block update use multipliers 0.1 and 0.05, with acceptance error at most $10^{-10} + 0.02$ times the absolute analytic derivative.

Positive-threshold brackets are eligible only if both block contributions exceed 10^{-10} in absolute magnitude, the finite-difference checks pass, and both bracket ratios lie in $[0.001, 1000]$. Noneligible combinations remain in the coverage denominator and retain the five fixed branches. Sign comparisons require both the predicted and observed change to exceed 10^{-10} in absolute magnitude; this is a numerical floor, not a practical-effect criterion. At five and twenty updates, the reported prediction is the frozen source tangent multiplied by the horizon, not a re-estimated or integrated along-path derivative. Validation labels enter threshold selection, so these are oracle diagnostics, not held-out confirmation of a learned rate-selection policy.

The imported direction summary reports all 27 planned state–stream jobs complete. Its counts and reported seed-level moments were checked locally; raw prediction and branch files were not available for independent inspection of finite-difference residuals or individual-stream effect sizes. Repeated streams are averaged within seed before reporting across-seed moments. Conditional threshold-bracket cells have differing coverage and must not be compared as complete unconditional seed groups. No new natural-image training was performed during manuscript revision.

F Proofs and Constants for Task-Relevant Motion

F.1 Regression, mobility, and near-conservation

Centering absorbs the affine intercept. Since $\mathbb{E}uu^\top = I$, minimizing over a decoder D_Y gives

$$\mathbb{E}\,\|Y - D_Y A u\|^2 = \operatorname{tr}\operatorname{Cov}(Y) - \|C_Y\|_F^2 + \left\|D_Y A - C_Y^\top\right\|_F^2. \tag{76}$$

Orthogonal projection onto row A proves Equation (11), including rank-deficient A. The differential formulas additionally require constant rank; the explicit bounds below use full row rank and a positive singular-value lower bound. Whitening the covariance support is exact, but estimating it from finite images is a separate statistical problem.

For a full-row-rank A, differentiating $P_A = A^\dagger A$ gives Equation (12). The two summands are adjoint off-diagonal blocks, whence

$$\left\|\dot{P}_A\right\|_F = \sqrt{2}\left\|A^\dagger \dot{A}(I - P_A)\right\|_F. \tag{77}$$

The head Jacobian is $\delta W \mapsto \delta W B$, with Gram action $G \mapsto GB^\top B$. A channel Jacobian has Gram T_c, so $\dot{B}_{c:} = -\eta_h (W^\top G)_{c:} T_c$. These facts prove Equation (13). Substituting $G = HWB + E$ and $T_c = \tau_c I + \mathcal{E}_c$, then multiplying by Q_B, eliminates the row-space term $\tau_c (U_0 B)_{c:}$ and proves Equation (18).

For linear V, $\dot{V} = -\eta_V W^\top H W V - \eta_V W^\top E$ proves Equation (14). Combining its exterior part with the projector identity yields the stated integral near-conservation bound. When $E = 0$, the continuous flow is invertible left multiplication on every finite bounded interval. In ordinary discrete gradient descent each step also left multiplies V; retaining its rank suffices for the same row space. Finite step size alone cannot produce drift under the exact isotropic gradient and rank preservation.

Differentiating GUPA with respect to $ASA^\top$ gives $G = AD - KAS$. Since $ASA^\top \succeq 0$, $\|K\|_2 \le 2$. Splitting S, D into scalar and residual terms proves Equation (17). This is conditional on a common clean/augmented moment support, as specified in the main text; it does not discard view-only coordinates.

F.2 The finite displacement bound

Put $P = P_0$, $Q = I - P$, and define

$$F = P(I - P_1)P, \quad E = QP_1Q, \quad K_1 = QP_1P. \tag{78}$$

The first two matrices are PSD. Equal ranks imply

$$\operatorname{tr} F = \operatorname{tr} E = d_{01}^2, \qquad \|K_1\|_F \le d_{01}. \tag{79}$$

For example, principal angles give $d_{01}^2 = \sum_i \sin^2\theta_i$ and $\|K_1\|_F^2 = \sum_i \sin^2\theta_i \cos^2\theta_i$. Exact block expansion gives

$$R(P_1) - R(P) = \operatorname{tr}(FPTP) - \operatorname{tr}(EQTQ) - 2\operatorname{tr}(K_1 PTQ). \tag{80}$$

Subtracting $\lambda_{\min}(T) I$ leaves the risk difference and task cross block unchanged. Each diagonal trace is then in $[0, \operatorname{osc}(T) d_{01}^2]$, so their difference has absolute value at most the upper endpoint. Cauchy–Schwarz bounds the cross term by $2a_0 d_{01}$, proving Equation (34). If $QTP = 0$ and the restricted row/normal spectra have separation δ, the same equality gives the one-sided bound δd_{01}^2.

Along a full-rank path, integrate the projector derivative to obtain

$$d_{01} \le \int \left\|A^\dagger \dot{A}(I - P_A)\right\|_F dt \le \frac{J}{\sigma}\int (\|\dot{h}\|_F + \|\dot{W}\|_F) dt. \tag{81}$$

This proves the finite update-budget bound. An invertible rescaling of representation coordinates leaves the risk fixed; a small singular value nevertheless makes a parameter-based sufficient bound more conservative. It does not itself prove information loss.

F.3 The complete two-block competition criterion

Let U, N be orthonormal bases of the current row and normal spaces. Every projector tangent has the form

$$D_g = UK_g^\top N^\top + NK_g U^\top. \tag{82}$$

For $\beta = Ux + Ny$, $\beta^\top D_g \beta = 2y^\top K_g x$. Thus absence of competition for every rank-one task is equivalent to

$$(y^\top K_h x)(y^\top K_W x) \ge 0 \quad \text{for all } x, y. \tag{83}$$

For fixed x with $K_h x \neq 0$, this requires $K_W x = \lambda(x) K_h x$ with $\lambda(x) \ge 0$: two noncollinear linear functionals admit opposite signs. If $z \in \ker K_h$, apply the same argument to $x + tz$ for every real t. A whole line cannot stay on one nonnegative ray unless $K_W z = 0$. Hence $K_W = F_0 K_h$ for a linear map F_0 on $\operatorname{im} K_h$. Every vector of that image is its eigenvector with a nonnegative eigenvalue. Linearity applied to two independent vectors and their sum makes that eigenvalue constant. Thus $K_W = \lambda K_h$, with $\lambda > 0$ if both tangents are nonzero. The zero-block cases and the converse are immediate. Positive proportionality also excludes competition for every PSD multi-output T.

Strict competition holds on an explicit task neighborhood. If a unit task β_0 has opposing contributions with minimum magnitude $m > 0$ and $d = \max_g \|D_g\|_2$, every unit task with $\|\beta - \beta_0\| \le m/(4d)$ retains both signs with margin at least $m/2$, since

$$|\beta^\top D_g \beta - \beta_0^\top D_g \beta_0| \le 2d\,\|\beta - \beta_0\|. \tag{84}$$

This states which task directions can compete at a fixed state, not the probability that a prescribed label task competes along random training.

F.4 Budget windows, finite steps, and explicit constants

For unit negative block directions, $\|d\theta/db\| \le 1$. On the ball in the main text, each $\langle \nabla R, u_g \rangle$ is L_q-Lipschitz. Consequently its deviation from its source value at budget s is at most $L_q s$. Integrating a convex combination proves Equation (38). For a constant allocation α, the sharper form is

$$|R_\alpha(b) - R_0 - b[\alpha q_h + (1-\alpha) q_W]| \le \tfrac{1}{2} L_q b^2. \tag{85}$$

Subtracting two such bounds proves the comparison bound. Source slopes separated from zero by m retain their signs with margin $m/2$ over the displayed half-window. For $q_h = -a < 0 < q_W = c$, both blocks can be active: allocations $\alpha_*/2$ and $(1+\alpha_*)/2$, where $\alpha_* = c/(a+c)$, have opposite slopes with margin at least $\min(a,c)/2$.

For perturbed Euler updates $\theta_{j+1} = \theta_j + \delta_j(F_\alpha(\theta_j) + e_j)$, $\|e_j\| \le \varepsilon$, and $b = \sum_j \delta_j$, assume $(1+\varepsilon)b \le \rho$. Taylor's theorem gives

$$\begin{aligned} |R_N - R_0 - b q_\alpha| \le{}& \tfrac{1}{2} L_q (1+\varepsilon) b^2 + G_R \varepsilon b \\ &+ \tfrac{1}{2} H_R (1+\varepsilon)^2 \sum_j \delta_j^2. \end{aligned} \tag{86}$$

The three terms bound state drift, direction error, and step curvature. Using $\sum \delta_j^2 \le \delta_{\max} b$ proves Equation (39) and its cumulative window. The perturbed process uses a nominal budget; its actual path length may differ. If raw block-gradient error is δ_g and its true norm is at least $\mu_g > 0$, normalized-direction error is at most $2\delta_g/\mu_g$. Thus a single batch does not automatically establish a useful uniform ε.

Here are fully explicit sufficient constants for the local square map. Let $k_\phi = \left\|U\Lambda^{1/2}\right\|_2$, and on a radius-$\rho$ parameter ball take $H_f = \|h_0\|_F + \rho$ and $W_f = \|W_0\|_2 + \rho$. Valid bounds are

$$J = k_\phi \sqrt{H_f^4 + 4W_f^2 H_f^2}, \tag{87}$$

$$H_A = 2k_\phi(H_f + W_f), \qquad a = k_\phi W_f H_f^2. \tag{88}$$

They bound DA, D^2A, and $\|A\|_F$, respectively. A sufficient rank bound is $\sigma = \sigma_{\min}(A_0) - J\rho > 0$. One may use

$$G_R = 2\|T\|_F J/\sigma, \tag{89}$$

$$H_R = 6\|T\|_F J^2/\sigma^2 + 2\|T\|_F H_A/\sigma. \tag{90}$$

For verification, $\nabla_A R = -2(A^\dagger)^\top T(I - P_A)$,

$$DA^\dagger[Z] = -A^\dagger Z A^\dagger + (I - P_A) Z^\top (AA^\top)^{-1}, \tag{91}$$

so $\left\|DA^\dagger[Z]\right\|_2 \le 2\|Z\|_F/\sigma^2$ and $\|DP_A[Z]\|_2 \le \|Z\|_F/\sigma$. Differentiation and composition give the displayed risk bounds.

For GUPA put $s_f = \|S\|_2$, $d_f = \|D\|_2$, $v_f = d_f + 2s_f$. The inverse resolvent bound yields

$$\|DG[Z]\|_F \le (v_f + 6s_f^2 a^2)\|Z\|_F. \tag{92}$$

Hence $H_L = (v_f + 6s_f^2 a^2)J^2 + a v_f H_A$ bounds the parameter-loss Hessian. If $\mu_g = \|g_g(\theta_0)\| - H_L\rho > 0$ for both blocks, take $K = H_L/\min_g \mu_g$. These constants require source moments and a specified region, rather than future fitting; they can be very conservative. They have not been shown useful for natural images.

For constant physical rates, ordinary gradient flow has budget allocation $\alpha(\theta) = \eta_h\|g_h\|/(\eta_h\|g_h\| + \eta_W\|g_W\|)$, which generally varies with state. Alternatively use clock $d\tau_{\text{rate}} = (\eta_h + \eta_W)dt$ and raw negative block fields. If their norms are bounded by M and Lipschitz constants by K_f, a corresponding Taylor remainder is $\frac{1}{2}(H_R M + G_R K_f) M \tau_{\text{rate}}^2$. A sum of rates is not a parameter path-length budget.

F.5 Gaussian patches and the two-filter motion criterion

For $x \sim N(0, I_p)$, Gaussian fourth moments make $u = \operatorname{svec}(xx^\top - I)/\sqrt{2}$ isotropic. Write $H_c = h_c h_c^\top$. Then $b_c = \sqrt{2}\operatorname{svec}(H_c)$ and

$$J_c v = \sqrt{2}\operatorname{svec}(h_c v^\top + v h_c^\top), \quad T_c[S] = 4(H_c S + S H_c). \tag{93}$$

For $G = HA$ with symmetric H, let $K_0 = W^\top H W$. The exact encoder update is

$$\dot{b}_c = -4\sqrt{2}\eta_h \operatorname{svec}\left(\sum_j (K_0)_{cj}(H_c H_j + H_j H_c)\right). \tag{94}$$

The Jordan product is $(h_c^\top h_j)(h_c h_j^\top + h_j h_c^\top)$. Orthogonal or collinear groups are a closure case: those products remain in the current span. For two independent filters put $\rho_h = h_1^\top h_2$, $n = Q_B \operatorname{svec}(h_1 h_2^\top + h_2 h_1^\top)$, and $p_B = B^\dagger(1,1)^\top$. Both n, p_B are nonzero, and

$$\dot{B} Q_B = -4\sqrt{2}\eta_h (K_0)_{12}\rho_h (1,1)^\top n^\top. \tag{95}$$

Thus, for $\eta_h > 0$, the encoder moves exactly when $(K_0)_{12}\rho_h \neq 0$. Its projector tangent is $vn^\top + nv^\top$ with $v = -4\sqrt{2}\eta_h(K_0)_{12}\rho_h p_B$. The unit tasks $(v/\|v\| \pm n/\|n\|)/\sqrt{2}$ have opposing risk derivatives of magnitude $\|v\|\,\|n\|$. A task exactly along n has zero first derivative and can only gain quadratically at the source. This establishes task relevance without asserting that all nonlinear encoders drift or that drift always hurts a task.

F.6 Proof of Theorem 1

In whitened input coordinates $q_c = \Sigma_x^{1/2} h_c$, Euclidean h training induces metric Σ_x on q. On the symmetric family $q_1 + q_2$ and $q_1 - q_2$ remain along e_+, e_-; both head

entries remain w. With $\kappa = 2\Phi'(V)$, direct differentiation gives

$$\dot{\lambda}_\pm = -8\eta_h \kappa w^2 \nu_\pm \lambda_\pm^2, \tag{96}$$

$$\dot{w} = -\eta_W \kappa w(\lambda_+^2 + \lambda_-^2). \tag{97}$$

Changing to τ gives Equation (30). Since $2\nu_+ > \nu_-$, the ratio λ_-/λ_+ strictly decreases. The unique state with $0 < \theta < \theta_0$ occurs at

$$\tau(\theta) = \frac{1/2 - \tan\theta}{\epsilon(\nu_+ - \nu_- \tan\theta)} < \frac{1}{2\nu_+\epsilon}. \tag{98}$$

The sum of the two encoder rows is the diagonal direction in the main text; their difference is a nonzero multiple of $E_\times$. This proves the ranks and Equation (32). At the two fixed comparison angles, the improvement and excess over initialization are respectively

$$(1 - z^2)\sin(\theta_0/4)\sin(7\theta_0/4 - 2\psi), \tag{99}$$

$$(1 - z^2)\sin(5\theta_0/6)\sin(2\psi - 7\theta_0/6). \tag{100}$$

Their minima on the task cone are the stated Δ_1, Δ_2. Independent additive label noise simply adds a common constant floor.

It remains to prove the descending regime and finite physical reachability rather than assume them. Population-centered Gaussian views with covariance-relative noise have $S = sI$, $D = 2(s-1)I$, $s = (1+\eta)^2 \le 36/25$. Thus $\kappa = -2\Gamma_s(sV)$ as in the original construction. Since $\theta_0 > 11/24$, $\tan\theta_2 > 3/40$. At θ_2 the closed formulas imply

$$\lambda_+ = \frac{2\epsilon(\nu_+/\tan\theta_2 - \nu_-)}{2\nu_+ - \nu_-} < 33\epsilon, \tag{101}$$

$$\lambda_- < \tfrac{5}{2}\epsilon, \qquad w^2 < \tfrac{1}{100} + \tfrac{5}{2}\frac{(65/2)\epsilon}{19/20} < \tfrac{87}{100}. \tag{102}$$

The modal values and w^2 increase up to this state, so throughout the interval

$$V = 2w^2(\lambda_+^2 + \lambda_-^2) < 2\tfrac{87}{100}\left[(33/100)^2 + (1/40)^2\right] < \tfrac{1}{5}. \tag{103}$$

Consequently $sV < 1/3$, and monotonicity of γ gives

$$\Gamma_s(sV) \ge 1 - \tfrac{36}{25}(1 - \gamma(1/3)) = \tfrac{101}{200} > \tfrac{1}{2}. \tag{104}$$

This strict barrier closes a first-exit argument for the descending regime. In particular $\dot{\tau} > 8\eta_h w_0^2$, while $\tau_2 < 1/(2\nu_+\epsilon) \le 10/(19\epsilon)$. Therefore $t_2 < 5/(76\eta_h w_0^2\epsilon)$. The exact formulas bound all parameters before this state, and nonzero front gradients make the loss strictly decrease, including when the head rate is zero.

The covariance interval is an explicit structured anisotropic family. It preserves the alignment and initialization conditions of the theorem. It does not supply a bound for arbitrary perturbations away from this invariant manifold. Similarly, the rate independence of the ordered subspace path is an exclusion result within this family, not a proof that rate bifurcation persists under symmetry breaking.

As an implementation check, `experiments/code/theory/analytic_bottleneck_check.py` evaluates 216 closed-form states, compares SVD risks on 25 tasks per state, and differentiates the Gaussian physical quadratic-moment objective in h, W for 648 state/noise combinations. The maximum risk and parameter-velocity discrepancies are about 1.25×10^{-15} and 1.44×10^{-15}. The checks include rotated covariance axes, preserve ranks two/one, and satisfy the uniform gaps. They do not replace the analytic proof or constitute random-initialization evidence. No trajectory is trained or integrated.

F.7 Origin-only interval windows on stored trajectories

The executable `experiments/code/theory/task_motion_check.py` uses the existing fixed-point interval and Taylor-jet arithmetic, but integrates no new trajectory. At each rational source state it constructs a positive forward parameter box D. Strictly positive vector-field components exclude exit through lower faces. If M_i bounds the upper component speed, $0.9\min_i(D_i^{\mathrm{upper}} - x_{0,i}^{\mathrm{upper}})/M_i$ is a valid conservative box lifetime. Fixed spectral powers are not dynamic coordinates. All rational denominators are bounded away from zero on the box.

Taylor jets on the whole box enclose the Lie second derivative $D(DRF)F$ by $[q_-, q_+]$; this includes vector-field acceleration, not just the risk Hessian. If $[d_-, d_+]$ encloses the source derivative, then

$$d_- t + \tfrac{1}{2}q_- t^2 \le R(t) - R_0 \le d_+ t + \tfrac{1}{2}q_+ t^2. \tag{105}$$

The chosen duration is the smaller of the box lifetime and half the ratio of the initial slope margin to the second-derivative bound. Integer endpoint checks verify the entire box inclusion and the half-margin inequality after rounding the candidate time downward. Four predetermined relative box widths, $1/100, 1/20, 1/10, 1/5$, are compared using source constants only; the saved trajectory file is opened after all predictions are computed.

The three scales $c = 1/2, 1, 3/2$ use the primary certificate center from Theorem 2, transformed by $W \mapsto c^{-2}W$, $B \mapsto c^2 B$. They share $A_0 = (0.02432, 0.02432)$ and initial risk $1/2$. Their certified physical windows exceed $0.06629, 0.09112, 0.03072$, respectively. Each contains saved future records; all five records satisfy the predicted amplitude envelopes. These are correlated deterministic trajectory records, not independent statistical trials.

On the same physical boxes the code encloses $0 < v_- \le \dot{\ell} \le v_+$. A first hit of $\ell_* = 1/50$ is guaranteed before the box expires when $\ell_*/v_- \le \tau$. Its time lies in $[\ell_*/v_+, \ell_*/v_-]$; substituting this interval into the preceding risk enclosure gives the common-energy predictions in Table 1. Direct differentiation in the energy clock is more conservative: its certified windows are only about

$0.01289, 0.01081, 0.02101$, and only the last directly covers one saved energy increment. The physical-box hitting-time argument is what certifies the common section for all three.

Mapping the scaled states back to the common original parameters gives rates (c^{-2}, c^4). After normalizing their sum to one, the three rate pairs are $(64/65, 1/65)$, $(1/2, 1/2)$, and $(64/793, 729/793)$. At common elapsed time $1/10$, source-only risk-change enclosures are conservatively

$$\begin{aligned} c = 1/2: &\quad [0.004267, 0.004917], \\ c = 1: &\quad [-0.002753, -0.002162], \\ c = 3/2: &\quad [-0.009328, -0.007343]. \end{aligned} \tag{106}$$

This is a finite split with the same initialization and rate sum. It does not equalize actual parameter path length. The common-energy comparison independently excludes a pure speed explanation. All interval constants and prediction rules precede reading the saved future values; the analysis itself is retrospective, not a new preregistered experiment. The CSV curves are floating-point numerical evidence, not newly certified integrations. The source-only certificates, rather than those curves, establish the displayed forward intervals. No natural-image or stochastic-gradient window is certified here.

G Natural-image protocols and detailed diagnostics

Natural-image experiments test whether direction selection has observable task consequences beyond the controlled construction. They reveal substantial objective improvement without substantial class-readability gains, and a local risk-direction effect with limited coverage and persistence. On CIFAR-10 [4], we use grayscale input, 64 11×11 valid convolutional filters, squaring, GAP, and an eight-dimensional head. Unsupervised pretraining starts from scratch without downstream labels. The training split is 45,000/5,000, and the standard 10,000-image test set is reserved for the fixed final-budget comparison. Augmentations are reflected crops, horizontal flips, contrast changes, and additive noise. Classification uses unregularized affine OLS on one-hot labels, evaluating encoder and projection separately. Validation MSE is a finite-sample task risk, not the theorem's population risk. Supervised references and the label-conditioned intervention below are distinguished from pretraining.

G.1 Layerwise rates do not establish a speed-independent gain

Five paired seeds and three ratios run for 4,000 updates at batch size 256, approximately 22.8 training-set passes. Encoder test accuracy is 36.48%, 36.45%, and 36.52% for $r = 0.1, 1, 10$; projection accuracy is 23.39%, 23.50%, and 23.81%. The fixed-budget projection difference is approximately 0.43 percentage points before rounding the group means. At common calibration energy $\ell = 2$, however, the corresponding validation difference is approximately $+0.11$ points with a five-seed descriptive bootstrap interval $[-0.42, +0.54]$. Other common-energy intervals also cross zero. Energy is measured on a fixed unlabeled calibration set and its first crossing is checked every ten updates; it is a proxy with recorded overshoot, not an exact population progress variable. The results do not establish either a stable speed-independent benefit or its absence.

Table 5: Validation OLS accuracy (%) for capacity diagnostics, $r = 1$; means over three development seeds. Each cell shows initialization $\rightarrow$ 4,000 updates.

Architecture	Encoder	Projection
$C = 64, K = 8$	$36.31 \rightarrow 36.04$	$23.49 \rightarrow 23.71$
$C = 256, K = 8$	$43.04 \rightarrow 43.04$	$23.66 \rightarrow 23.65$
$C = 64, K = 32$	$36.31 \rightarrow 36.23$	$32.05 \rightarrow 32.14$

G.2 More capacity mostly improves the initial features

Development diagnostics use seeds 0, 1, and 12, three architectures, and two supervised references, for 33 runs of 4,000 updates. These runs use validation labels only for evaluation, never the test set. Table 5 separates initial performance from learned improvement.

Width and projection dimension improve absolute readability, but their gains largely exist at random initialization. Supervised training of the original model reaches encoder OLS accuracy 40.07% through the encoder path and projection OLS accuracy 38.69% through the joint projection path. Thus 24% is not an expressivity ceiling for eight-dimensional outputs. The supervised references use Adam and prescribed gradient clipping, unlike unsupervised SGD, and are attainable references rather than matched optimization or capacity upper bounds.

The wide model has a small exploratory ratio signal: $r = 10$ minus $r = 0.1$ projection accuracy is positive for all three development seeds at $\ell = 1$ and 2, with means near $+0.20$ and $+0.19$ points. Multiple inspected sections, few development seeds, and inconsistent MSE signs at some sections prevent a stable intervention claim. Some $K = 32$ energy sections are missing; we retain those missing cases rather than treating available pairs as a complete experiment.

G.3 A fixed encoder admits a much better eight-dimensional head

We freeze the original $r = 1$ encoder at initialization and update 4,000 for each development seed: six source states, with only three independent seeds. Each state

Table 6: Same frozen encoder, different heads. Validation accuracy (%), three paired development seeds per source state.

Readout	Initial	Trained
Full encoder OLS	36.31	36.04
Original head + OLS	23.49	23.71
Rank-8 least squares	35.75	35.59
CE-trained head + OLS	30.89	28.49
CE-trained classifier	30.83	27.73

compares the original head, full encoder OLS, an optimal rank-8 affine least-squares fit, and a supervised head trained with cross-entropy (CE). The rank-8 solution uses training labels only. Its affine intercept is absorbed into the final decoder, leaving the projection itself linear. CE training updates only the projection and classifier; kernels are checked for exact equality before and after training.

For the trained encoder, rank-8 fitting improves accuracy over the original head by approximately 11.88 points, remaining only 0.45 points below full encoder OLS. Its validation one-hot MSE is 0.08147432, compared with 0.08631537 for the original head and 0.08141745 for the full encoder. It removes 98.84% of this particular MSE gap; this is not a percentage of all semantic information. A better compressed choice exists without any encoder change.

Crucially, almost the same gap is already present at initialization. These data do not show that pretraining created a large new semantic loss. The CE-trained head improves over the source head but falls short of rank-8 fitting even on training accuracy (28.71% versus 36.59%). The gap is therefore not solely a validation overfitting effect, but CE versus square loss, augmented versus clean inputs, scaling, clipping, and budget are confounded; factorized optimization alone is not identified as its cause.

Ten-class centered one-hot targets have at most nine independent directions, so near-recovery by eight coordinates has a direct algebraic context. Reduced-rank fitting is globally optimal for training MSE, not validation accuracy. It establishes available task-dependent compression, not a label-free way to find it or an architecture-specific phenomenon.

G.4 Objective improvement amplifies existing image statistics

The architecture reads local second moments: for valid image patches p_u and filter h_c, its encoder coordinate is $h_c^\top M(x) h_c$, where $M(x) = |U|^{-1} \sum_u p_u p_u^\top$. This makes existing image-energy statistics plausible optimization directions, but does not establish that they are unrelated to classes.

We examine this explanation on the Describable Textures Dataset (DTD), using official partition 1, 64×64 grayscale inputs, the same 64/8 model, and three development seeds (0, 1, 12), with $r = 1$ and 2,000 updates. OLS fits use the official training set and evaluation uses the validation set; no test images are evaluated. In the original pilot, the encoder's leading clean-feature variance share rises from 91.93% to 99.46%, while the tail variance remains nonzero. A train-fitted predictor using the squared per-image DC offset relative to the training global mean explains approximately 93.4% of the final leading-component variation on validation images. Encoder training accuracy changes from 17.82% to 17.70%, and validation accuracy from 8.65% to 8.51%. Thus finite-label probe generalization is a limitation, but cannot alone explain the absence of training-set class gains.

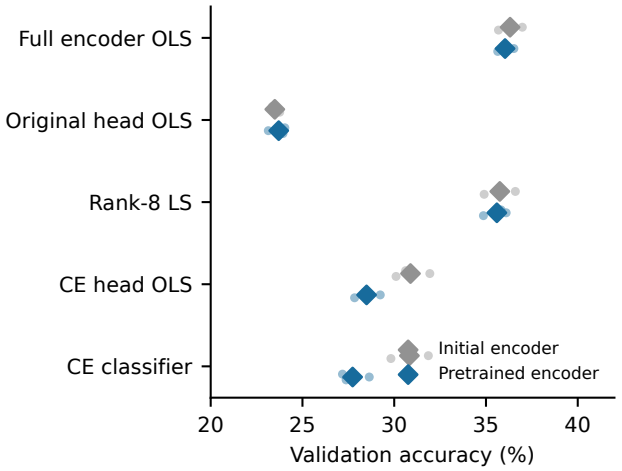


Figure 4: Validation accuracy for different readouts of the same fixed encoder. Points are paired development seeds. The large source-head gap exists both before and after pretraining. Rank-8 fitting is supervised.

Removing each filter's spatial mean at evaluation suppresses the identified DC response without reliably improving the class task. Training a spatially zero-mean model from initialization likewise produces no clear encoder learning gain: validation accuracy changes from 8.78% to 8.56%. Its projection validation MSE improves slightly, from 0.02072149 to 0.02071484, while accuracy changes from 6.22% to 6.21%. The dominant direction now tracks pixel variance, explaining about 84% of final projection PC1 validation variation; this association already exists at initialization. The projection remainder has a small positive signal (accuracy 4.65% → 4.89%), but does not establish improved full-encoder readability. The zero-DC constraint changes the initial function and training dynamics, so the two runs are not matched-function interventions.

Together these observations are consistent with preferential amplification of existing, augmentation-stable second-order statistics, rather than substantial acquisition of class-predictive directions. They do not isolate the objective, augmentation, or architecture as the unique

Table 7: CIFAR threshold brackets: observed opposite risk signs that both match the source-state prediction. Coverage is 4/27 state–stream combinations across three development seeds. These are repeated measurements, not independent trials.

Branch updates	1	5	20
Successful brackets / eligible	4/4	2/4	0/4

cause. Spectral concentration alone is not evidence of task-information deletion: full-rank unregularized OLS is invariant to invertible feature-coordinate transformations, and the recorded tail variance and numerical rank persist. Conversely, the observed non-scalar feature changes do not by themselves establish task-relevant learning.

G.5 Local risk reversal exists but has limited persistence

We directly test two-block competition using the CIFAR $r = 1$ checkpoints at updates 0, 1,000, and 4,000 for the same three development seeds. Each state has three paired future streams, giving 27 state–stream combinations, not 27 independent samples. Let R denote validation one-hot MSE after refitting training-set affine OLS. With source mobility m, base rate η, and block training gradients g_h, g_W, define

$$A = \langle \nabla_h R, -\eta m g_h \rangle, \qquad B = \langle \nabla_W R, -\eta m g_W \rangle,$$
$$\Delta R(r, s) \approx \frac{2s}{1+r}(A + rB). \tag{107}$$

The derivative includes the training decoder's dependence on the features. Unlike the analytic population threshold, this is an empirical, label-conditioned calculation. A positive threshold $r_* = -A/B$ requires opposing block contributions. We compare fixed ratios $0.1, 1, 10$, overall speed controls $s = 0.5, 2$ at $r = 1$, and eligible brackets $r_*/2, 2r_*$. Predictions precede branch evaluation; implementation and eligibility details are in Appendix E.

All 27 combinations pass the recorded finite-difference checks. Only four have eligible positive thresholds: none of nine at initialization, one of nine at update 1,000, and three of nine at update 4,000. All four brackets reverse the observed risk sign as predicted after one update, but only two retain that reversal after five updates and none after twenty (Table 7). The first-step changes are tiny: available seed-level means within source-state/bracket cells have absolute magnitude 7.49×10^{-10} to 4.71×10^{-9}, with no accuracy change. First-step agreement checks the local approximation; it is not by itself evidence of a useful control rule or a mechanism unique to this architecture.

A small fixed-ratio effect remains at the late source state: after twenty updates, $r = 10$ minus $r = 1$ validation MSE is -3.72×10^{-8}, -2.74×10^{-7}, and -2.41×10^{-8} for seeds 0, 1, and 12, respectively, averaging streams within each seed. This is improvement relative to the baseline branch, not uniform improvement relative to the source: seed 1 still increases its risk. Each seed's stream-mean accuracy change for $r = 10$ is zero.

The results establish a local correspondence with two-block direction competition, but not the certified finite-time rebound or a stable natural-image intervention. Beyond the first update, both batches and parameters change. Loss of the source threshold's predictive control therefore does not contradict Equation (107); the present measurements do not separate stochastic-gradient variation from state-dependent changes in the local contributions.